Deneb

MYMENSINGH GIRLS' CADET COLLEGE
PROGRESS TEST EXAMINATION - 2025
CLASS: HSC
MULTIPLE CHOICE QUESTIONS
STATISTICS
SECOND PAPER
[According to the Syllabus of 2026]
TIME – 20 minutes
FULL MARKS – 20

Subject Code: | 1 | **3** | 0 |

Set: **Ka**

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the "Answer sheet" for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. $P(A \cup B) = P(A) + P(B)$ **implies A & B are –**

   (a) Disjoint (b) Independent (c) Joint (d) Independent

   Answer the next THREE questions based on the following information.

   A card is drawn at random from a standard deck of 52 playing cards.

2. **What is the probability that the card is a Queen?**

   (a) 0.0192 (b) 0.0769 (c) 0.25 (d) 0.5

3. **P(The card is not a Spade)**

   (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$

4. **P(The card is black or a Heart)**

   (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

5. **In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected. What is the probability that the student passed in at least one subject**

   (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{4}{5}$

6. $P(A) = \frac{2}{3}, P(B) = \frac{1}{5},$ **and** $P(A \cap B) = \frac{1}{6}$

   i. A and B are independent
   ii. $P(A \cup B) = \frac{7}{10}$
   iii. $P(\bar{A}) = \frac{1}{3}$

   Which one is correct?

   (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

   Answer the next three questions based on the following information

X	1	2	3
P(x)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

7. **What is** $P(X \geq 2)$**?**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

8. **Find the cumulative probability** $F(2)$**.**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

9. $P(X = 1$ **or** $X = 3)$ **equals**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

10. **For a discrete random variable** X **with probability mass function** $p(x)$**:**

    i. $P(a \leq X \leq b) = \sum_{x=a}^{b} p(x)$
    ii. $\sum_x p(x) = 1$
    iii. $p(x) \geq 0$ for all x

    Which one is correct?

    (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

    Answer the next TWO questions based on the following information

    Let X be a continuous random variable uniformly distributed on the interval $[2, 6]$.

11. **What is $E(X)$?**

(a) 2 (b) 3 (c) 4 (d) 5

12. **What is $V(X)$?**

(a) $\frac{4}{3}$ (b) $\frac{8}{3}$ (c) $\frac{16}{3}$ (d) $\frac{5}{3}$

Answer the next TWO questions based on the following information

X	1	2	3
P(x)	0.5	0.3	0.2

13. **What is $E(X^2)$?**

(a) 3.6 (b) 4.2 (c) 5.0 (d) 5.6

14. **What is $V(2X)$?**

(a) 2.24 (b) 3.36 (c) 4.48 (d) 5.60

Answer the next TWO questions based on the following information

X	2	4	6
P(x)	0.5	0.3	0.2

15. **What is $E(X)$?**

(a) 2.5 (b) 3.4 (c) 4.0 (d) 4.6

16. **What is $V(X)$?**

(a) 2.04 (b) 2.44 (c) 2.84 (d) 3.24

Answer the next TWO questions based on the following information

X	0	1	4
P(x)	0.4	0.4	0.2

17. **What is $E(X^2)$?**

(a) 1.2 (b) 2.0 (c) 3.6 (d) 4.8

18. **What is $V(X)$?**

(a) 2.24 (b) 2.64 (c) 3.04 (d) 3.44

Answer the next TWO questions based on the following information

The probability mass function of a discrete random variable X is given by

$$P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

19. **What is $E(X)$?**

(a) 2.0 (b) 2.5 (c) 3.0 (d) 3.5

20. **What is $V(X)$?**

(a) 0.50 (b) 0.75 (c) 1.00 (d) 1.25

Answer the next TWO questions based on the following information

The probability mass function of a discrete random variable X is given by

$$P(X = x) = \frac{x}{6}, \quad x = 1, 2, 3.$$

21. **What is $E(X)$?**

(a) 1.33 (b) 2.00 (c) 2.33 (d) 3.00

22. **What is $V(X)$?**

(a) 0.33 (b) 0.56 (c) 0.89 (d) 1.11